

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS

WEDNESDAY, 8 OCTOBER 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS**Question 1**

- a) What are the TWO (2) differences between **normal array** and **vector array**? Show an example in C++ on how each of these types of arrays are declared for *scores* of integer type. (6 marks)
- b) In reference to an integer type vector array **numbers**, write C++ statements using vector related functions to perform the following tasks:
- (i) Add 23 to the **numbers** array. (2 marks)
 - (ii) Remove the second element from the **numbers** array. (2 marks)
- c) Write C++ program statements to perform the following tasks: -
- (i) Declare a two-dimensional array of double type called **sales** for a store to record the daily sales of a week. Its row and column represent 7 days of a week, and the 10 products sold in the store respectively.
Note that the order for the week being recorded in the array starts from Monday and the product is being recorded as product 1 to product 10. (2 marks)
 - (ii) Calculate and display the total sales made on Friday. (7 marks)
- d) (i) Determine the bit pattern of 1000 1001 after right shift 3 bits. (1 mark)
- (ii) Determine the bit pattern of X where 1101 0011 & X produce 0101 0010. (1 mark)
- e) Explain the difference between Functional Cohesion and Coincidental Cohesion. (2 marks)
- f) Given the following recursive function, identify the *base case* and explain the purpose of having the base case.

```
void countdown(int seconds)
{
    if (seconds == 0) {
        cout << "Time's up!" << endl;
        return;
    }
    cout << seconds << " seconds left..." << endl;
    countdown(seconds - 1); // recursive call
}
```

(2 marks)

[Total: 25 marks]

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTSQuestion 2

- a) (i) Explain the string function **std** and show an example in C++ on how it is used. (2 marks)
- (ii) What is the answer of the following C++ program segment? Give a reason for the answer.
- ```
string text = "abc123?";
regex pattern("[a-z]+\d+");
if(regex_match(text, pattern))
 cout << "Full match!";
else
 cout << "No match!";
```
- (2 marks)
- b) Write C++ statements to accomplish the following tasks:
- (i) Define a structure **Mark** which consists of integer type members for the subject's **math**, **science** and **English**. Each of these subjects stores marks for TEN(10) students. (3 marks)
- (ii) Declare a variable **mark** of Mark structure type. (2 marks)
- (iii) Prompt the users to enter the marks for all the subjects for the ten students. (6 marks)
- (iv) Replace the mark for **Science** of the 7<sup>th</sup> student with 95. (2 marks)
- (v) Define a structure **Faculty** with members **facultyName** and **deptName**. (2 marks)
- (vi) Rewrite the **Mark** structure by adding in a nested structure member called **faculty** of **Faculty** structure type. (2 marks)
- (vii) Declare a structure variable **marks** of type **Mark**. (1 mark)
- (viii) Change the first letter of the **deptName** to 'S'. (3 marks)

[Total: 25 marks]

Question 3

- a) What is the main difference between a C++ function using *reference parameter* and another function using *return statement* in returning any output value from the function. (2 marks)
- b) Develop a C++ program segment for the following tasks: -
- (i) Declare a boolean type C++ function **analyseNum** which receives an integer **num**, two integer type parameters **square** and **cube** which are using pass by reference and pass by address respectively. The function will calculate the square ( $\text{num} * \text{num}$ ) and cube ( $\text{num} * \text{num} * \text{num}$ ). It will also check whether the num is even. The function should return true if num is even, and false otherwise. (8 marks)

**AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS****Question 3 b) (Continued)**

- (ii) Write a **main** function to prompt the user to enter an integer. Then, call the **analyseNum** function to calculate and display the square and cube of the integer. The function should also display whether the integer is an even number or not. The sample screen is shown below: -

Enter an integer: 5

Result

Square of the integer: 25

Cube of the integer: 125

Even number: false

(9 marks)

- c) Examine the output of the following C++ program. Note that the memory address of **p**, **q** and **r** are OE89XC0, OE89XC4, and OE89XC8 respectively.

```
#include <iostream>
using namespace std;
int main() {
 int p = 20, q, r;
 int *m, *n;
 m = &p;
 n = &q;
 *n = 88;
 r = *m + q;
 cout << "p = " << *m << endl;
 cout << "q = " << q << endl;
 cout << "r = " << r << endl;
 m = &r;
 (*n)++;
 p = *m - *n;
 cout << "p = " << p << endl;
 cout << "q = " << q << endl;
 cout << "&m = " << m << endl;
 return 0;
}
```

(6marks)

[Total: 25 marks]

**Question 4**

- a) Write C++ statements for an enumerated type **Cafe** to represent the following values and then declare a variable **favouriteCoffee** which is initialised with "Cotti".

Zus Coffee, San Francisco Coffee, Gigi, Cotti, HWC, Starbuck

(4 marks)

**AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS****Question 4 (Continued)**

- b) You are building a simple function for a calculator that can add two values and return the result. Prepare a *template* in C++ for this calculator function so that it is reusable for adding values of different types. (4 marks)
- c) Perform the following tasks using C++ statements:
- (i) An integer type **score** array is used to store the scores of a test taken by 20 students in a class. Write a function **checkCount** which receives the **score** array and a selected score of a student. The function will find and display the count of scores of all students that are lower than the selected score. Below is the sample message displayed:
- “The selected score 90 is higher than the scores of 8 students in the class.”
- (9 marks)
- (ii) Write a function call statement to call the above **checkCount** function where it will determine the score of the last student in the score array is higher than how many students in the class. (2 marks)
- d) Given the following structure declaration for Employee: -
- ```
struct Employee {  
    string name, id;  
    double salary;  
};
```
- (i) Develop a C++ function **updateSalary** to perform a 5% salary increment exercise in a company where the function receives the employee record as per the above structure and update the salary by applying the increment. The calling function must be able to access the updated salary value from the employee record. (4 marks)
- (ii) Construct a function calling C++ statement to call the **updateSalary** function developed in Question 4 d) (i). Assume structure variable **emp** of type Employee was declared. (2 marks)

[Total: 25 marks]